

DHGD: Dynamic Hand Gesture Dataset for Skeleton-Based Gesture Recognition and Baseline Evaluations

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Abstract: In this paper, we introduce the Dynamic Hand Gesture Dataset (DHGD) tailored for skeleton-based gesture recognition. This novel dataset encompasses nine distinct hand gestures, each performed by twenty participants, designed to facilitate intuitive manipulation of images, including actions such as rotations and scaling. Each gesture sequence is captured over 70 frames using an RGB-D camera system. Key points of the hand are detected in each frame utilizing MediaPipe Hands, with key point represented by its three-dimensional coordinates. Additionally, we evaluate the recognition accuracy of the nine distinct dynamic hand gestures using Spatial-Temporal Graph Convolutional Network (ST-GCN). This network is selected as a benchmark for graph convolutional networks to validate the dataset's efficacy.

I. INTRODUCTION

Hand gesture recognition has emerged as a pivotal touchless user interface, gaining widespread adoption in gaming, entertainment, and medical applications, as evidenced by our touchless visualization system [1]. Recent advancements have leveraged multimodal data, such as color and depth images [2-7], bolstering the precision of gesture recognition system. While static gesture recognition has attained commendable accuracy, it fails to capture the subtle dynamics of human interactions, consequently restricting its utility in real-time and dynamic contexts. To address this deficiency, dynamic gesture recognition has become imperative, offering a more authentic and intuitive user experience by incorporating temporal dynamics and the fluidity of natural motion overlooked by static models. In this paper, we have developed DHGD, a novel dataset for dynamic hand gesture recognition utilizing skeleton-based approaches and deep learning. The

primary benefits of this dataset are outlined as follows:

- A diverse collection of nine distinct hand gestures, demonstrated by a group of thirty participants, resulting in a rich and diversified dataset
- The dynamic hand gestures are specifically designed for natural and intuitive interaction with visual interfaces.
- A benchmark analysis employing the Spatial-Temporal Graph Convolutional Network (ST-GCN) has been executed on the dataset. Detailed recognition accuracies for each unique gesture are provided, which assist users in identifying the most effective dynamic gestures for their applications.

The remainder of this paper is organized as follows: Section 2 describes the dataset's unique attributes in detail. Section 3 discussed the evaluation of the dataset against benchmark performance metrics. Finally, Section 4 presents our concluding observation and implications of the research findings.

II. SKELTON GRAGH DYNAMIC HAND GESTURE DATASETS

A. Ergonomic Design of the Dataset Recording System

During the experimental procedure, participants were positioned 0.9 meters away from the sensor, which was utilized to record the gestures using a single RGB-D camera, specifically the Kinect [8]. Figure 1 schematically depicts the setup for hand gesture capture, with the hand images being recorded against a blue sheet that served as the standardized background.

B. Hand Image Pre-processing

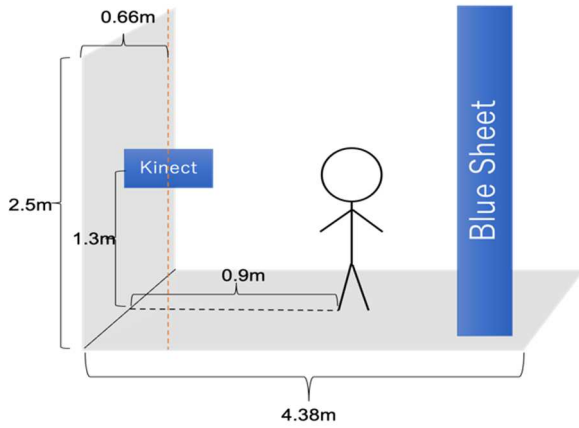


Fig.1 Illustration of the DHGD dataset acquisition setup.

The hand color images were synthesized from the skeleton tracking and depth data provided by the Kinect system. The initial step involved capturing raw color images of the participant's hand, as depicted in Figure 2(a). A meticulous calibration procedure was then implemented to ensure precise alignment between the color and depth sensors, with a particular focus on the articulation points of the right hand. For targeted analysis, we demarcated a 300×300 pixel square as the Region of Interest (ROI), as shown in Figure 2(b). This defined ROI allows for a concentrated examination of the gesture dynamics within a consistent frame of reference.

Upon completion of the data collection phase, we utilized MediaPipe Hands [9] for the detection of joint points. The output data includes coordinates of hand landmarks in both image and world reference frames, as well as the classification of hand orientation, distinguishing between left and right hands. Figure 2(c) exemplifies the imagery post-application of MediaPipe Hands, and Figure 2(d) exhibits the reconstructed 3D visualization derived from the output.

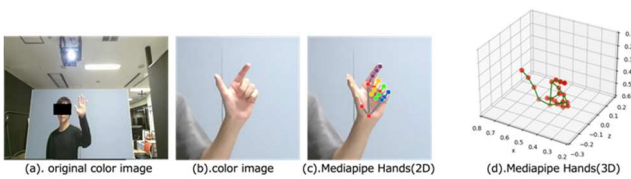
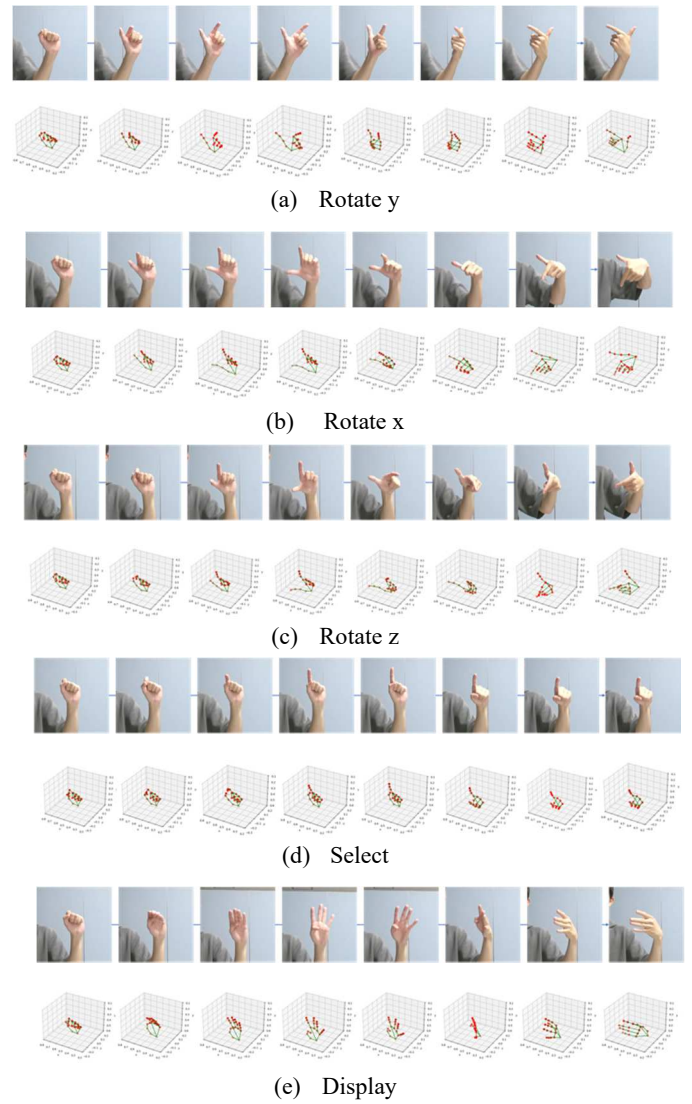


Fig.2. Sample data for hand image pre-processing

C. Dataset Characteristics

This dataset comprises nine distinct hand gestures, each captured in a sequence of 70 frames to comprise a complete motion sequence. Recorded from thirty participants, every individual performed each gesture five times, culminating in a comprehensive dataset of 900 instances ($30 \text{ participants} \times 9 \text{ gestures} \times 5 \text{ repetitions}$). The resolution for each hand gesture image is standardized at 300 by 300 pixels, ensuring uniformity in the visual data. Significantly, the dataset includes nine intuitive hand gestures that mimic common rotational and scaling action. Figures 3 (a to h) illustrate the

diverse range of gestures included in the dataset. Gestures (a to c) are predicated on rotational dynamics, with the thumb and index finger articulating around the x, y, and z axes, respectively. Gesture (d) involves the elevation of the index finger followed by a downward motion, mimicking a pressing action. Gesture 3(e) is characterized by displaying four fingers and then drawing them inward while concurrently tilting the palm sideways. In Gesture 3(f), the motion starts with fingers directed downwards and culminates in their upward elevation. Conversely, Gesture 3(g) depicts the inverse action, where fingers begin elevated and are subsequently lowered. Gestures (h) and (i) represent the actions of expansion and contraction, respectively, where the deliberate spreading and narrowing of the thumb and index finger exemplify these concepts.



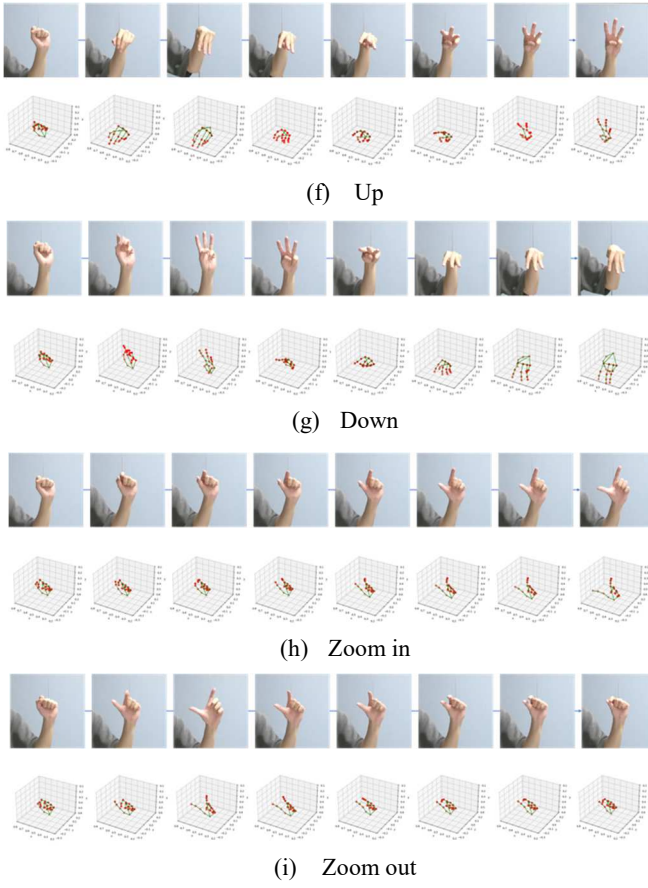


Fig. 3 Nine dynamic hand gestures

III. BENCHMARK EVALUATION

A. Baseline techniques and configurations

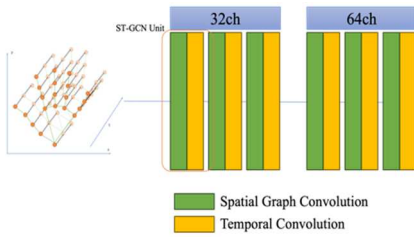


Fig.4. ST-GCN model for hand gesture recognition

Table.1. Recognition accuracy by ST-GCN for each gesture

Gesture	ST-GCN [%]	ST-GCN+DA [%]
y	87.78	93.61
x	86.94	89.45
z	78.89	81.11
select	96.39	93.33
surface	99.45	99.45
up	99.45	99.45
down	98.34	98.89
zoom in	90.56	97.78
zoom out	94.17	96.11
total	92.41	94.35

The Spatial-Temporal Graph Convolutional Network (ST-GCN) [10] is a model introduced in 2018 to detect human actions using skeletal data. This model comprises multiple layers of spatial and temporal graph convolutions, aimed at learning spatial and temporal graphs from the input data. ST-GCN initiates with spatial convolution, extending the GCN model into the temporal domain. Subsequently, convolution occurs across frames, advancing in the temporal direction through iterative application.

By employing convolution in both temporal and spatial dimensions, ST-GCN effectively captures temporal sequence information. These elements are denoted as ST-GCN Units. The architecture encompasses six layers of ST-GCN units, subsequently followed by a Global Average Pooling (GAP) layer and a Softmax classifier. The configuration of the ST-GCN model employed for hand gesture recognition is illustrated in Figure 4.

B. Evaluation Results

To ensure robust validation, a eight-fold cross-validation approach was employed. In this scheme, data from 21 subjects were designated for training, while the data from 9 subjects were reserved for testing, resulting in a total of 945 training samples and 405 test samples encompassing all nine classes. This validation process was reiterated times, and the outcomes consistently affirmed in each iteration. The average classification accuracy achieved using ST-GCN is detailed in Table 1, summarizing the recognition accuracy for each gesture. Additionally, we experimented by randomly translating coordinates for the training data at each epoch and applying data augmentation (DA). While the recognition accuracy for rotation and zooming in and out appears relatively lower compared to other gestures, this discrepancy can be attributed to significant individual variations. Nonetheless, the overall recognition accuracy for the test set reached 92.41%.

Furthermore, applying data augmentation to the test data resulted in a 1.94% improvement in accuracy, achieving a recognition accuracy of 94.35%. The confusion matrix is illustrated in Figure 5.

Gesture	y	x	z	Sel	Dis	Up	Down	in	out
Rorate y	93.61	0.56	5.83	0.00	0.00	0.00	0.00	0.00	0.00
Rorate x	0.00	89.45	9.72	0.00	0.00	0.00	0.83	0.00	0.00
Rorate z	4.44	9.17	81.14	0.00	0.56	0.28	0.00	0.28	1.95
Select	0.00	1.11	0.28	93.33	0.00	0.00	1.39	2.78	1.11
Display	0.00	0.00	0.28	0.00	99.45	0.00	0.00	0.28	0.00
Up	0.00	0.00	0.28	0.00	0.28	99.45	0.00	0.00	0.00
Down	0.00	0.00	0.28	0.00	0.63	0.28	98.89	0.00	0.00
Zoom in	0.00	0.00	0.28	1.39	0.00	0.28	0.00	97.78	0.28
Zoom out	0.56	0.00	0.83	0.83	0.00	0.00	0.00	1.67	96.11

Fig.5. ST-GCN + DA confusion matrix

IV. CONCLUSION

In this paper, we presented a dynamic hand gesture dataset for skeleton-based gesture recognition based on Kinect sensor and MediaPipe Hands. A comprehensive collection of 9 dynamic hand gestures was captured from 30 participants and subsequently subjected to meticulous preprocessing to extract well-segmented hand region images. The ST-GCN model was employed as a benchmark for this dataset, demonstrating an accuracy of 94.35%. Future efforts will be dedicated to further enhancing the accuracy of dynamic hand gesture recognition, with the overarching objective of fully unleashing the potential of the Skeleton Graph Dynamic Hand Gesture Datasets.

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